Crush Syndrome

TB-03 | Trauma & Burns | Reviewed

Clinical release pending. Verify current approved protocols and standing orders.

Quick flow

01 Entrapment / Compression [EMT / AEMT / Paramedic]

Confirm scene safety, duration, body region, access, and rescue plan.

02 Crush Syndrome Risk? [EMT / AEMT / Paramedic]

Large muscle mass • any significant entrapment time • swelling • neurologic deficit • shock. Document the time compression began when known.

03 Before Release [AEMT / Paramedic]

Airway • continuous cardiac monitoring/12-lead when feasible • IV/IO • normal saline preferred • prepare defibrillator and request early Medical Control/rescue-physician input. ECG-directed calcium, bicarbonate, and high-dose albuterol are Paramedic-only. Do not delay unsafe extrication.

04 Release + Reassess [EMT / AEMT / Paramedic]

Coordinate release with rescue personnel; assign a clinician to monitor the patient. Watch immediately for hypotension, hyperkalemia/dysrhythmia, respiratory failure, arrest, and compartment syndrome.

05 Treat Complications [AEMT / Paramedic]

Reassess shock, pain, compartment syndrome, and associated trauma; supportive treatment within scope. Request Paramedic treatment for suspected hyperkalemia.

06 Paramedic Hyperkalemia Care [Paramedic]

Prepare ECG-directed calcium, sodium bicarbonate, and high-dose albuterol treatment before release for ECG changes, dysrhythmia, or peri-arrest deterioration under the reconciled standing order or Medical Control. No routine prophylactic calcium or bicarbonate.

07 Rapid Trauma Transport [EMT / AEMT / Paramedic]

Minimize scene delay once safely released and notify the trauma center early for high-risk entrapment.

Overview

- Suspect crush syndrome after compression of a large muscle mass, especially an extremity or torso; it can occur in under an hour and risk rises with longer entrapment.
- Begin resuscitation before release when feasible because reperfusion can cause sudden hyperkalemia, acidosis, shock, and cardiac arrest.

Provider-Level Actions

Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT

- Coordinate safe rescue, control bleeding, manage airway and ventilation, prevent hypothermia, splint as indicated, and document the compression time and body region.
- Do not delay extrication for nonessential procedures when the scene is unsafe.

AEMT

- Perform all EMT care plus large-bore vascular access and 0.9% saline before release when feasible: adult initial 1 L, then continue reassessed fluids during extrication. Prefer normal saline, but do not withhold otherwise indicated isotonic crystalloid solely because saline is unavailable. Use Medical Control for pediatric fluid guidance and for additional fluid rate/reassessment limits not specified by standing order.

- Do not administer calcium, sodium bicarbonate, or high-dose albuterol for hyperkalemia as an AEMT; these interventions are Paramedic-only.
- Monitor rhythm and perfusion, obtain a 12-lead when feasible, and prepare for rapid deterioration at the moment of release.

Paramedic

- Perform all prior care plus protocol-directed treatment for ECG changes, dysrhythmia, or peri-arrest hyperkalemia, severe pain, and shock; do not give routine prophylactic calcium or bicarbonate solely for crush injury.
- Coordinate the timing of release with rescue personnel and ensure the defibrillator and ECG-directed hyperkalemia treatment are immediately available. Obtain early Medical Control/rescue-physician input for prolonged large-muscle compression and follow the high-risk crush-release plan.
- Pre-release empiric medication use, fluid rate, and reassessment limits require the specified standing order or Medical Control; do not improvise. Do not repeat calcium gluconate without a specific Medical Control order confirming the cumulative dose.

Indications

- Patient with suspected crush syndrome requiring prehospital assessment or treatment.

Warnings / precautions

- Do not apply a prophylactic tourniquet solely because crush syndrome is suspected. Consider one only for hemorrhage control or with Medical Control/an established rescue plan when pre-release treatment cannot be provided.
- Reperfusion-related deterioration may occur immediately after the compressive force is removed.

Source and references

Claiborne County EMS native protocol content; controlled PDF generated from the same structured source. | Current Tennessee EMS scope of practice, verified individual credentials, and Claiborne County standing orders control provider authorization.

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